



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CO1_AT3_Visualization Design Exercise

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DSA0601- Data Handling and Visualization

Question 1

A multinational retail company has collected annual sales information from its branches across different regions. Using the given dataset, design an effective visualization that simultaneously represents **Sales, Profit, Number of Customers, Product Category, and Region**. Select suitable aesthetic mappings and apply an appropriate scale transformation wherever necessary. Justify every visualization design choice.

R Dataset retail <- data.frame(

Branch=c("B1","B2","B3","B4","B5","B6","B7","B8","B9","B10",

"B11","B12","B13","B14","B15","B16","B17","B18","B19","B20"),

Region=c("North","South","East","West","North","South","East","West

", "North","South","East","West","North","South","East","West",

"North","South","East","West"),

Category=c("Electronics","Furniture","Clothing","Food","Electronics",

"Furniture","Clothing","Food","Electronics","Furniture",

"Clothing","Food","Electronics","Furniture","Clothing",

"Food","Electronics","Furniture","Clothing","Food"),

Sales=c(450000,620000,180000,920000,350000,720000,250000,1080000,

560000,670000,195000,980000,430000,760000,280000,1150000,

510000,690000,220000,1250000),

Profit=c(45000,95000,-12000,185000,25000,120000,18000,225000,

8000,205000,51000,132000,24000,250000,

69000,115000,15000,285000),

72000,98000,-

```
Customers=c(1200,2500,980,4200,1800,3000,1250,5200,
2100,3300,1100,4700,1750,3600,1450,5600,
2200,3900,1600,6100)
)
```

Aim

Design a bubble chart to visualize Sales, Profit, Customers, Product Category, and Region of retail branches using **ggplot2**.

Explanation:

1. The bubble chart displays Sales and Profit for each retail branch.
2. Bubble size represents the number of customers.
3. Different colours represent different regions.
4. Different shapes represent different product categories.
5. This visualization makes it easy to compare branch performance and customer distribution.

R Code

```
# Install packages (Run only once)
install.packages("ggplot2") install.packages("scales")

# Load libraries
library(ggplot2) library(scales)
#
Dataset
retail <- data.frame(
  Branch=c("B1", "B2", "B3", "B4", "B5", "B6", "B7", "B8", "B9", "B10",
"B11", "B12", "B13", "B14", "B15", "B16", "B17", "B18", "B19", "B20"),
  Region=c("North", "South", "East", "West", "North", "South", "East", "West"
,
"North", "South", "East", "West", "North", "South", "East", "West",
"North", "South", "East", "West"),
  Category=c("Electronics", "Furniture", "Clothing", "Food", "Electronics"
,
"Furniture", "Clothing", "Food", "Electronics", "Furniture",
"Clothing", "Food", "Electronics", "Furniture", "Clothing",
"Food", "Electronics", "Furniture", "Clothing", "Food"),
  Sales=c(450000, 620000, 180000, 920000, 350000, 720000, 250000, 1080000,
560000, 670000, 195000, 980000, 430000, 760000, 280000, 1150000,
510000, 690000, 220000, 1250000),
```

```

Profit=c(45000,95000,-12000,185000,25000,120000,18000,225000,
        72000,98000,-8000,205000,51000,132000,24000,250000,
        69000,115000,15000,285000),
Customers=c(1200,2500,980,4200,1800,3000,1250,5200,
            2100,3300,1100,4700,1750,3600,1450,5600,
            2200,3900,1600,6100)
)

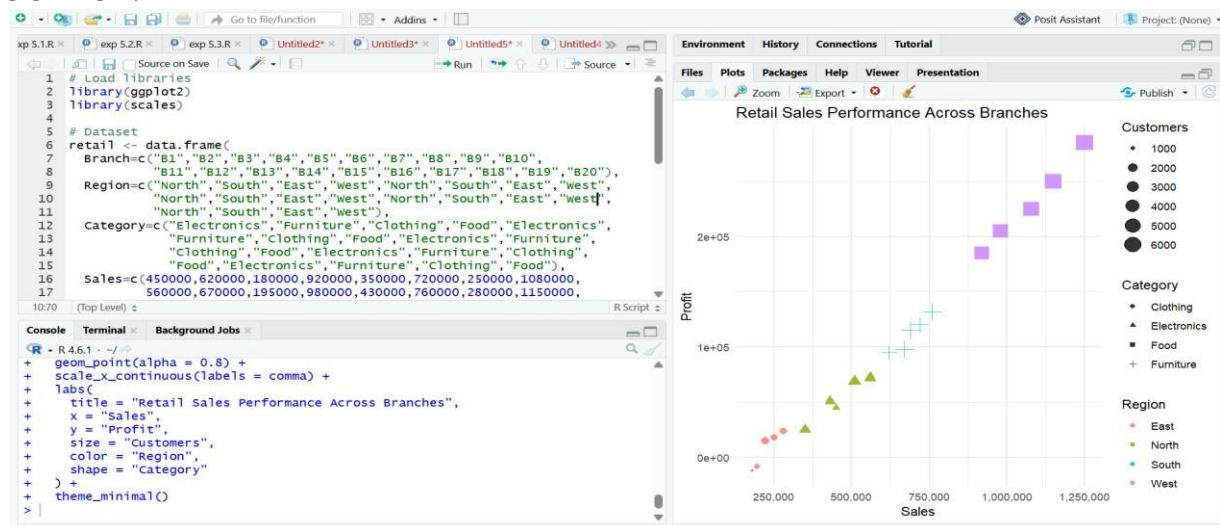
```

```

# Bubble Chart ggplot(retail,
aes(x = Sales,          y = Profit,
size = Customers,      color =
Region,                shape = Category))
+   geom_point(alpha = 0.8) +
scale_x_continuous(labels = comma) +
labs(
    title = "Retail Sales Performance Across Branches",
x = "Sales",    y = "Profit",    size = "Customers",
color = "Region",    shape = "Category"
) +
theme_minimal()

```

OUTPUT:



Question 2

The following dataset contains monthly electricity generation from different energy sources. Develop visualizations using **Cartesian Coordinates** and **Polar Coordinates**. Compare both visualizations and determine which coordinate system provides better analytical insight. Explain your design decisions.

R Dataset

```
energy <- data.frame(
```

```
Month=month.abb,
```

```
Coal=c(620,580,590,605,640,670,690,720,700,680,650,630),
```

```
Gas=c(310,295,305,320,340,355,360,370,365,350,335,320),
```

```
Solar=c(45,60,90,130,180,230,250,245,210,160,90,50),
```

```
Wind=c(80,95,120,150,170,185,195,190,175,150,110,90),
```

```
Hydro=c(140,150,160,170,180,195,205,210,200,185,165,150)
```

```
)
```

Aim

Develop Cartesian and Polar coordinate visualizations to compare monthly electricity generation from different energy sources.

Explanation

1. The Cartesian plot shows electricity generation trends over the months.
2. The Polar plot displays the same data in a circular format.
3. Different colours represent different energy sources.
4. The Cartesian plot is better for comparing monthly values accurately.
5. The Polar plot is useful for showing seasonal or cyclic patterns.

R Code

```
# Install packages (Run only once)
install.packages("ggplot2") install.packages("reshape2")

# Load libraries library(ggplot2)
library(reshape2)
# Dataset energy <-
data.frame(
  Month = month.abb,
  Coal = c(620,580,590,605,640,670,690,720,700,680,650,630),
  Gas = c(310,295,305,320,340,355,360,370,365,350,335,320),
  Solar = c(45,60,90,130,180,230,250,245,210,160,90,50),
  Wind = c(80,95,120,150,170,185,195,190,175,150,110,90),
  Hydro = c(140,150,160,170,180,195,205,210,200,185,165,150)
)

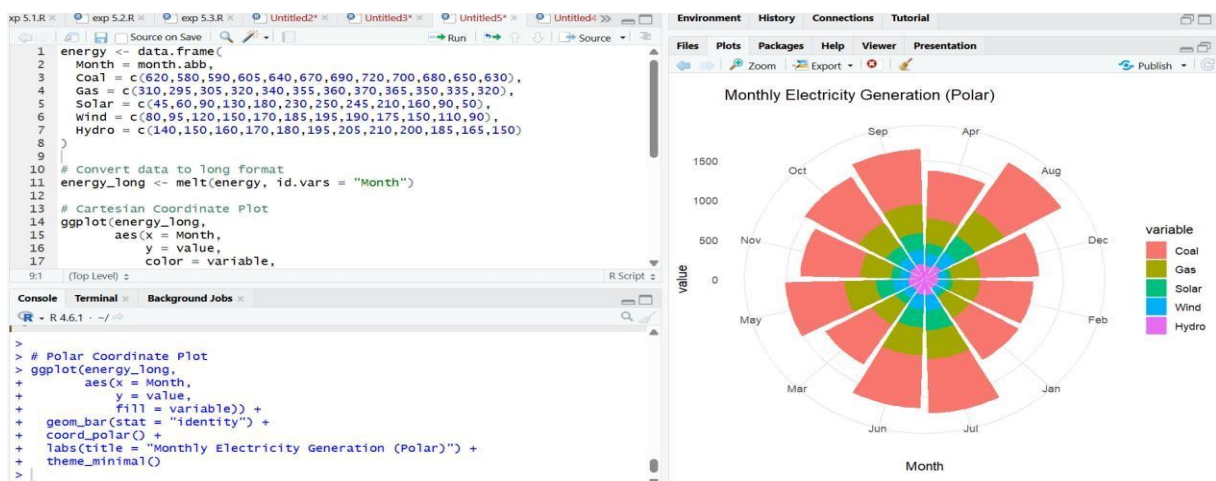
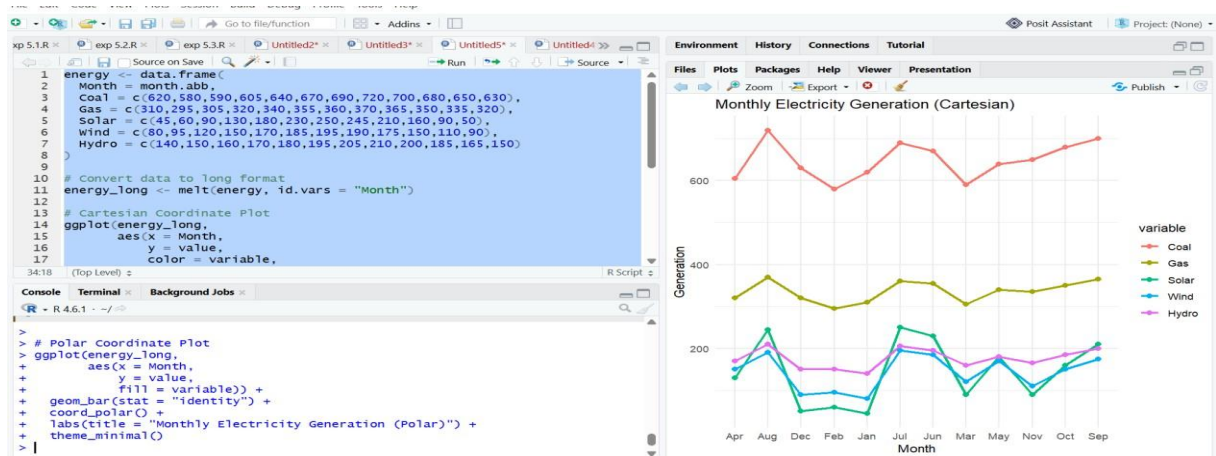
# Convert data to long format
```

```
energy_long <- melt(energy, id.vars = "Month")

# Cartesian Coordinate Plot
ggplot(energy_long,
aes(x = Month,
    y = value,
    color = variable,
    group = variable)) + geom_line(size = 1) + geom_point(size = 2) +
  labs(title = "Monthly Electricity Generation (Cartesian)",
x = "Month",
    y = "Generation") + theme_minimal()

# Polar Coordinate Plot
ggplot(energy_long,
aes(x = Month,
    y = value,
    fill = variable)) + geom_bar(stat = "identity") + coord_polar() +
  labs(title = "Monthly Electricity Generation (Polar)") + theme_minimal()
```

Output



Question 3

An environmental monitoring agency has collected climate data from major Indian cities.

Design a visualization that effectively represents **Temperature, Air Quality Index (AQI), Rainfall, and Humidity** using appropriate colour scales. Select a suitable colour palette and justify its effectiveness for accurate interpretation and accessibility. **R Dataset** climate <- data.frame(

City=c("Delhi","Mumbai","Chennai","Kolkata","Hyderabad",

"Bangalore","Pune","Ahmedabad","Jaipur","Lucknow",

"Bhopal","Patna","Goa","Kochi","Nagpur","Indore",

"Surat","Visakhapatnam","Coimbatore","Mysuru"),

Temperature=c(44,35,38,40,39,31,33,43,45,39,

38,40,32,30,41,37,36,34,29,28),

Humidity=c(22,84,78,72,48,66,58,26,18,42,

36,45,89,91,30,34,76,82,68,64),

Rainfall=c(6,280,220,190,85,120,95,2,1,35,

48,72,360,410,12,20,180,240,130,145),

AQI=c(365,110,125,180,160,72,84,290,320,240,

175,220,58,52,270,190,145,98,60,55)

)

Aim

Design a visualization to represent Temperature, Humidity, Rainfall, and Air Quality Index (AQI) of Indian cities using appropriate colour scales.

Explanation

1. The bubble chart compares Temperature and AQI for different cities.
2. Bubble size represents the amount of rainfall.
3. Bubble colour represents humidity levels.
4. The colour gradient helps distinguish low and high humidity.
5. This visualization makes climate conditions easy to compare across cities.

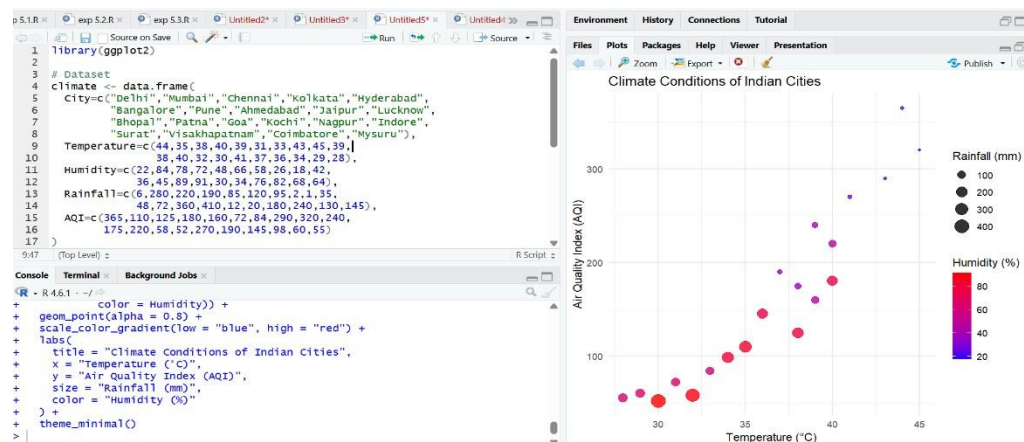
R Code

```
# Install package (Run only once) install.packages("ggplot2")

# Load library library(ggplot2)
#
Dataset
climate <- data.frame(
  City=c("Delhi", "Mumbai", "Chennai", "Kolkata", "Hyderabad",
        "Bangalore", "Pune", "Ahmedabad", "Jaipur", "Lucknow",
        "Bhopal", "Patna", "Goa", "Kochi", "Nagpur", "Indore",
        "Surat", "Visakhapatnam", "Coimbatore", "Mysuru"),
  Temperature=c(44, 35, 38, 40, 39, 31, 33, 43, 45, 39,
               38, 40, 32, 30, 41, 37, 36, 34, 29, 28),
  Humidity=c(22, 84, 78, 72, 48, 66, 58, 26, 18, 42,
            36, 45, 89, 91, 30, 34, 76, 82, 68, 64),
  Rainfall=c(6, 280, 220, 190, 85, 120, 95, 2, 1, 35,
            48, 72, 360, 410, 12, 20, 180, 240, 130, 145),
  AQI=c(365, 110, 125, 180, 160, 72, 84, 290, 320, 240,
       175, 220, 58, 52, 270, 190, 145, 98, 60, 55)
)

# Bubble Chart ggplot(climate,
aes(x = Temperature,
y = AQI,
Rainfall,
Humidity)) + geom_point(alpha
= 0.8) +
  scale_color_gradient(low = "blue", high = "red") +
labs(
  title = "Climate Conditions of Indian Cities",
  x = "Temperature (°C)", y = "Air Quality Index
(AQI)", size = "Rainfall (mm)", color =
"Humidity (%)"
) +
theme_minimal()
```

Output



Question 4

A software company has recorded project statistics for twenty software development teams.

Design a visualization that requires the use of a **nonlinear axis transformation** to represent highly skewed data. Justify the selected transformation and explain how it improves interpretation over a linear scale. **R Dataset** software <- data.frame(

Team=paste("T",1:20,sep=""),

LinesOfCode=c(12000,18000,25000,35000,48000,

62000,81000,96000,120000,150000,

185000,240000,310000,390000,480000,

620000,810000,1050000,1350000,1800000),

Bugs=c(35,52,60,72,88,

110,125,142,165,188,

205,240,280,320,365,

420,490,560,640,720),

Developers=c(5,7,9,10,12,

14,16,18,20,22,

25,28,30,34,38,

42,46,50,55,60),

Complexity=c(2.1,2.4,2.7,3.0,3.2,

3.5,3.8,4.1,4.5,4.8,

5.1,5.4,5.8,6.2,6.6,

7.1,7.6,8.0,8.5,9.0)

)

Aim

Design a visualization using a nonlinear (logarithmic) axis transformation to represent highly skewed software project data.

Explanation

1. The scatter plot compares Lines of Code and Number of Bugs.
2. Bubble size represents the number of developers.
3. Bubble colour represents project complexity.
4. A logarithmic scale is used because the Lines of Code values are highly skewed.
5. The log scale improves readability and makes comparisons easier.

R Code

```
# Install package (Run only once) install.packages("ggplot2")

# Load library library(ggplot2)
#
Dataset
software <- data.frame(  Team =
paste("T",1:20,sep=""),
  LinesOfCode = c(12000,18000,25000,35000,48000,
                  62000,81000,96000,120000,150000,
                  185000,240000,310000,390000,480000,
                  620000,810000,1050000,1350000,1800000),
  Bugs = c(35,52,60,72,88,
           110,125,142,165,188,
           205,240,280,320,365,
           420,490,560,640,720),
  Developers = c(5,7,9,10,12,
                14,16,18,20,22,
                25,28,30,34,38,
```

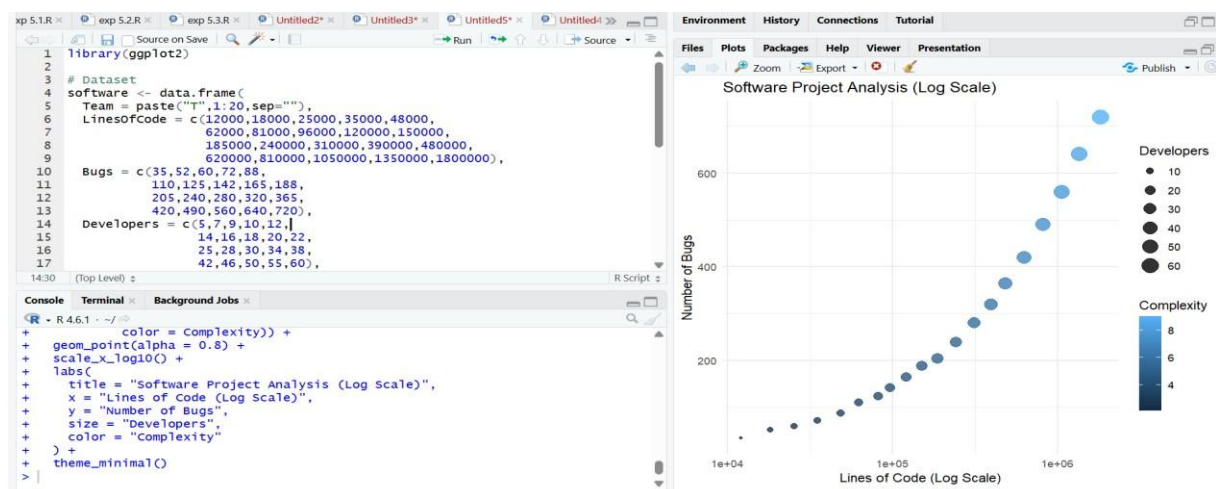
```

        42,46,50,55,60),
  Complexity = c(2.1,2.4,2.7,3.0,3.2,
                3.5,3.8,4.1,4.5,4.8,
                5.1,5.4,5.8,6.2,6.6,
                7.1,7.6,8.0,8.5,9.0)
)

# Scatter Plot with Log Scale
ggplot(software, aes(x =
LinesOfCode, y = Bugs,
size = Developers,
color = Complexity)) +
geom_point(alpha = 0.8) +
scale_x_log10() + labs(
  title = "Software Project Analysis (Log Scale)",
  x = "Lines of Code (Log Scale)", y = "Number of
Bugs", size = "Developers", color =
"Complexity"
) +
theme_minimal()

```

Output



Question 5

A financial analytics company has collected performance data for twenty publicly listed companies. Develop an integrated visualization that effectively communicates **Revenue, Profit Margin, Market Capitalization, Employee Strength, and Industry Sector**. Use appropriate aesthetic mappings, coordinate systems, scales, and colour palettes to maximise interpretability while minimising perceptual bias. **R Dataset** companies <- data.frame(

```

Company=paste("C",1:20,sep=""),

Sector=c("IT","Banking","Healthcare","Retail","Energy",

"IT","Banking","Healthcare","Retail","Energy",

"IT","Banking","Healthcare","Retail","Energy",

"IT","Banking","Healthcare","Retail","Energy"),

Revenue=c(420,580,720,850,980,

1150,1320,1480,1650,1820,

2050,2280,2550,2840,3150,

3480,3860,4250,4680,5200),

ProfitMargin=c(8.2,15.6,11.4,9.3,18.5,

20.4,17.8,14.2,10.5,22.1,

24.5,19.8,16.7,13.4,26.2,

28.1,21.5,18.3,15.2,30.4),

Employees=c(1500,3200,2100,4200,2800

, 5100,6800,3900,7200,4500,

8100,9500,6200,10300,7200,

11800,13200,8600,14500,16000),

MarketCap=c(8500,12000,9800,7600,14500

, 18200,21500,16800,13200,24500,

28500,32500,24800,19200,38200,

42500,46800,35600,29800,52000))

```

Aim

To design an integrated bubble chart that represents Revenue, Profit Margin, Market Capitalization, Employee Strength, and Industry Sector using appropriate aesthetic mappings, colour scales, and coordinate systems.

Explanation

1. The bubble chart compares Revenue and Profit Margin.
2. Bubble size represents Market Capitalization.
3. Different colours represent different industry sectors.
4. The chart helps compare company performance across sectors.
5. The integrated visualization displays multiple financial measures in a single graph.

R Code

```
# Install package (Run only once) install.packages("ggplot2")

# Load library library(ggplot2)
#
Dataset
companies <- data.frame(
  Company = paste("C",1:20,sep=""),
  Sector = c("IT","Banking","Healthcare","Retail","Energy",
             "IT","Banking","Healthcare","Retail","Energy",
             "IT","Banking","Healthcare","Retail","Energy",
             "IT","Banking","Healthcare","Retail","Energy"),
  Revenue = c(420,580,720,850,980,
              1150,1320,1480,1650,1820,
              2050,2280,2550,2840,3150,
              3480,3860,4250,4680,5200),
  ProfitMargin = c(8.2,15.6,11.4,9.3,18.5,
                  20.4,17.8,14.2,10.5,22.1,
                  24.5,19.8,16.7,13.4,26.2,
                  28.1,21.5,18.3,15.2,30.4),
  Employees = c(1500,3200,2100,4200,2800,
                5100,6800,3900,7200,4500,
                8100,9500,6200,10300,7200,
                11800,13200,8600,14500,16000),
  MarketCap = c(8500,12000,9800,7600,14500,
                18200,21500,16800,13200,24500,
                28500,32500,24800,19200,38200,
                42500,46800,35600,29800,52000)
)

# Bubble Chart
ggplot(companies,
  aes(x = Revenue, y = ProfitMargin,
      size = MarketCap,
      color = Sector)) +
  geom_point(alpha = 0.8) +
  labs(
    title = "Company Financial Performance",
    x = "Revenue", y = "Profit Margin (%)",
```

```

size = "Market Capitalization",      color =
"Industry Sector"
) +
theme_minimal()

```

Output:

